

1. main.c

```
#include "header.h"
```

```
#include <stdio.h>
```

```
int main(){
```

```
    int questionNumber;
```

```
    printf("Enter Question Number ( 1 - 15 ):");
```

```
    scanf("%d",&questionNumber);
```

```
    switch (questionNumber)
```

```
{
```

```
case 1: {
```

```
    // Q1
```

```
    int size = 10;
```

```
    int arr1[size];
```

```
    int i, answer;
```

```
    printf("Enter %d numbers:\n", size);
```

```
    for(i = 0 ; i < size ; i++){
```

```
        scanf("%d",&arr1[i]);
```

```
    }printf("Given numbers are : \n");
```

```
    for(i = 0 ; i < size ; i++){
```

```
        printf("%d ",arr1[i]);
```

```
    }
```

```
    answer = SumOfSquares(arr1 , size);
```

```
    printf("\n");
```

```
    printf("The Sum of Squares of given %d numbers is %d\n",size ,answer);
```

```
    break;
```

```
}
```

```
case 2: {
```

```
    // Q2
```

```
    int size = 10;
```

```

int arr2[size];

printf("Enter %d numbers:\n", size);

for(int i = 0 ; i < size ; i++){
    scanf("%d",&arr2[i]);
}

printf("Reverse of Given Array of Integers is : \n");

ReverseOfArray(arr2 , size);

printf("\n");

break;
}

case 3: {
    // Q3
    int size = 50;
    float arr3[size];
    float target;
    printf("Enter %d numbers:\n", size);
    for(int i = 0 ; i < 50 ; i++){
        scanf("%f",&arr3[i]);
    }

    printf("Enter Target\n");
    scanf("%f",&target);

    int answer = FindingElement(arr3 , 50 , target);

    if(answer != 0){
        printf("Index of Given Target Number is %d\n",answer);
    }

    else{
        printf("Not Found\n");
    }

    break;
}

case 4: {

```

```

// Q4
int arr4[] = {60 , 700 , 80 , 900 , 10};
int size = sizeof(arr4)/sizeof(arr4[0]);
TriangularPattern(arr4 , size);
break;
}
case 5: {
    // Q5
    int size;

    printf("Enter size of integer array in bytes : ");
    scanf("%d",&size);

    printf("Number of Elements in array is %d",NumberOfElements(size));
    break;
}
case 6: {
    // Q6
    int arr5[] = {70 , 80 , 90 , 100 , 110};
    printf("Original Array: ");
    for(int i = 0; i < 5; i++){
        printf("%d ",arr5[i]);
    }
    printf("\n");
    int size = sizeof(arr5)/sizeof(arr5[0]);
    int n;
    printf("Enter no of rotations : ");
    scanf("%d",&n);
    RightShift(arr5 , size , n);
    printf("Right Shift with %d rotations : ",n);
    for(int i = 0 ; i < size ; i++){
        printf("%d ",arr5[i]);
    }
}

```

```

int arr6[] = {70 , 80 , 90 , 100 , 110};

printf("\n");

LeftShift(arr6 , size , n);

printf("Left Shift with %d rotations : ",n);

for(int i = 0 ; i < size ; i++){

    printf("%d ",arr6[i]);

}

printf("\n");

break;

}

case 7: {

    // Q7

    int arr7[] = {40 , 50 , 50 , 50 , 60 , 70 , 80 , 90 , 60 , 100 , 10};

    printf("Original Array: ");

    for(int i = 0; i < 11; i++){

        printf("%d ",arr7[i]);

    }

    printf("\n");

    int size = sizeof(arr7)/sizeof(int);

    printf("Array after Deleting Duplicate Elements : ");

    DeleteDuplicate(arr7 , size);

    break;

}

case 8: {

    // Q8

    printf("Ten random numbers in [1,100]\n");

    RandomNoGenerator(1 , 100 , 10);

    break;

}

case 9: {

```

```

// Q9

int arr9[20];

int size = 20;

printf("Enter %d numbers:\n", size);

for(int i = 0 ; i < 20 ; i++){

    scanf("%d",&arr9[i]);

}

printf("Given Numbers are : \n");

for(int i = 0 ; i < 20 ; i++){

    printf("%d ",arr9[i]);

}

printf("\n");

NoOfPosNumbers(arr9 , size);

NoOfNegNumbers(arr9 , size);

NoOfOddNumbers(arr9 , size);

NoOfEvenNumbers(arr9 , size);

NoOfZeroes(arr9 , size);

break;

}

case 10: {

    // Q10

    int arr10[] = {3 , 6 , 0 , 6 , 3};

    int size = sizeof(arr10)/sizeof(arr10[0]);

    printf("Given Numbers are : \n");

    for(int i = 0 ; i < size ; i++){

        printf("%d ",arr10[i]);

    }

    printf("\n");

    Palindrome(arr10 , size);

    break;

}

```

```

case 11: {
    // Q11
    int arr11[] = {10, 45 , 32 , 16 , 88};
    printf("Original Array: ");
    for(int i = 0; i < 5; i++){
        printf("%d ",arr11[i]);
    }
    printf("\n");
    int size = sizeof(arr11)/sizeof(arr11[0]);
    ReverseArray(arr11 , size);
    printf("Reverse of Given Array is : ");
    for(int i = 0 ; i < size ; i++){
        printf("%d ",arr11[i]);
    }
    break;
}

case 12: {
    // Q12
    int arr12[] = {1 , 2 , 3 , 4 , 5 , 6 , 11 , 8 , 12 , 10};
    printf("Original Array: ");
    for(int i = 0; i < 10; i++){
        printf("%d ",arr12[i]);
    }
    printf("\n");
    int size = sizeof(arr12)/sizeof(arr12[0]);
    int target;
    printf("Enter Target Number : \n");
    scanf("%d",&target);
    printf("Nearest Lesser : %d\n",NearestLesser(arr12 , size , target));
    printf("Nearest Greater : %d\n",NearestGreater(arr12 , size , target));
    break;
}

```

```
}
```

```
case 13: {
```

```
    // Q13
```

```
    int A[] = { 45, 50, 70, 85, 90};
```

```
    int B[] = { 30, 40, 60, 75, 80};
```

```
    int C[10];
```

```
    SortMixArray(A , B , C);
```

```
    printf("Sorted Mixed Array is : \n");
```

```
    for(int i = 0 ; i < 10 ; i++){
```

```
        printf("%d ",C[i]);
```

```
    }
```

```
    break;
```

```
}
```

```
case 14: {
```

```
    // Q14
```

```
    int arr14[1000];
```

```
    int newarr[1000];
```

```
    GenerateArray(arr14);
```

```
    printf("Random Generated Array is : ");
```

```
    for (int i = 0; i < 1000; i++) {
```

```
        printf("%d ", arr14[i]);
```

```
    }
```

```
    printf("\n");
```

```
    printf("Numbers which are divisible by 8 or 25 in the array are : \n");
```

```
    int size = NewArray(arr14 , newarr);
```

```
    for (int i = 0 ; i < size ; i++){
```

```
        printf("%d ", newarr[i]);
```

```
    }
```

```
    break;
```

```
}
```

```
case 15: {  
    // Q15  
    int arr15[] = {1 , 2 , 3 , 4 , 5 , 6 , 7};  
    printf("Original Array: ");  
    for(int i = 0; i < 7; i++){  
        printf("%d ",arr15[i]);  
    }  
    printf("\n");  
    int size = sizeof(arr15)/sizeof(arr15[0]);  
    int secondLargest = SecondLargest(arr15 , size);  
    printf("Second Largest Element in the Given Array is : %d",secondLargest);  
    break;  
}  
default:  
    break;  
}  
return 0;  
}
```


2. header.h

```
#ifndef HEADER_H
#define HEADER_H

int SumOfSquares(int arr[] , int size);
void ReverseOfArray(int arr[] , int size);
int FindingElement(float arr[] , int size , float target);
void TriangularPattern(int arr[] , int size);
int NumberOfElements(int size);
void RightShift(int arr[] , int size, int n);
void LeftShift(int arr[] , int size , int n);
void DeleteDuplicate(int arr[] , int size);
void RandomNoGenerator(int min , int max , int no);
void NoOfPosNumbers(int arr[] , int size);
void NoOfNegNumbers(int arr[] , int size);
void NoOfOddNumbers(int arr[] , int size);
void NoOfEvenNumbers(int arr[] , int size);
void NoOfZeroes(int arr[] , int size);
void Palindrome(int arr[] , int size);
void ReverseArray(int arr[] , int size);
int NearestLesser(int arr[] , int size , int target);
int NearestGreater(int arr[] , int size , int target);
void SortMixArray(int A[] , int B[] , int C[] );
void GenerateArray(int arr[]);
int NewArray(int arr[] , int newarr[]);
int SecondLargest(int arr[] , int size);
#endif // HEADER_H
```

3. logic.c

```
#include "header.h"
```

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
#include <stdlib.h>
```

```
// Q1
```

```
int SumOfSquares(int arr[], int size){
```

```
    int i , sum = 0;
```

```
    for(i = 0 ; i < size ; i++){
```

```
        sum += (arr[i]*arr[i]);
```

```
    }
```

```
    return sum;
```

```
}
```

```
// Q2
```

```
void ReverseOfArray(int arr[] , int size){
```

```
    int i;
```

```
    for(i = size-1 ; i >= 0 ; i--){
```

```
        printf("%d ",arr[i]);
```

```
    }
```

```
    return;
```

```
}
```

```
// Q3
```

```
int FindingElement(float arr[] , int size , float target){
```

```
    int i;
```

```
    for(i = 0 ; i < size ; i++){
```

```
        if(arr[i] == target){
```

```
            return i;
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```

// Q4

```
void TriangularPattern(int arr[] , int size){
```

```
    int i,j;
```

```
    for(i = 0 ; i < size ; i++){
```

```
        for(j = 0 ; j <= i ; j++){
```

```
            printf("%d ",arr[j]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

// Q5

```
int NumberOfElements(int size){
```

```
    int number = size/sizeof(int);
```

```
    return number;
```

```
}
```

// Q6

```
void RightShift(int arr[] , int size, int n){
```

```
    for(int i = 1 ; i <= n ; i++){
```

```
        int temp = arr[size-1];
```

```
        for(int j = size-1 ; j >= 0 ; j-- ){
```

```
            arr[j] = arr[j - 1];
```

```
        }
```

```
        arr[0] = temp;
```

```
    }
```

```
}
```

```
void LeftShift(int arr[] , int size , int n){
```

```
    for(int i = 1 ; i <= n ; i++){
```

```
        int temp = arr[0];
```

```
        for(int j = 0 ; j < size - 1 ; j++){
```

```
            arr[j] = arr[j + 1];
```

```
        }
```

```
    arr[size-1] = temp;
}
}
```

// Q7

```
void DeleteDuplicate(int arr[] , int size){
    for(int i = 0 ; i < size ; i++){
        for(int j = i+1 ; j < size ; j++){
            if(arr[i] == arr[j]){
                for(int k = j ; k < size - 1 ; k++){
                    arr[k] = arr[k+1];
                }
                size--;
                j--;
            }
        }
    }
    for(int i = 0 ; i < size ; i++){
        printf("%d " , arr[i]);
    }
}
```

// Q8

```
void RandomNoGenerator(int min , int max , int no){
    int c, n;
    for (c = 1; c <= no; c++) {
        n = rand() % (max - min + 1) + min;
        printf("%d\n", n);
    }
}
```

// Q9

```
void NoOfPosNumbers(int arr[] , int size){  
    int count = 0;  
    for(int i = 0 ; i < size ; i++){  
        if(arr[i] > 0){  
            count++;  
        }  
    }  
    printf("Number of positive integers in the array is %d\n",count);  
}
```

```
void NoOfNegNumbers(int arr[] , int size){  
    int count = 0;  
    for(int i = 0 ; i < size ; i++){  
        if(arr[i] < 0){  
            count++;  
        }  
    }  
    printf("Number of negative integers in the array is %d\n",count);  
}
```

```
void NoOfOddNumbers(int arr[] , int size){  
    int count = 0;  
    for(int i = 0 ; i < size ; i++){  
        if(arr[i] % 2 != 0){  
            count++;  
        }  
    }  
    printf("Number of odd integers in the array is %d\n",count);  
}
```

```
void NoOfEvenNumbers(int arr[] , int size){  
    int count = 0;  
    for(int i = 0 ; i < size ; i++){
```

```

        if(arr[i] % 2 == 0){
            count++;
        }
    }

    printf("Number of even integers in the array is %d\n",count);
}

```

```

void NoOfZeroes(int arr[] , int size){
    int count = 0;
    for(int i = 0 ; i < size ; i++){
        if(arr[i] == 0){
            count++;
        }
    }

    printf("Number of zeroes in the array is %d\n",count);
}

```

// Q10

```

void Palindrome(int arr[] , int size){
    int flag = 0;
    for(int i = 0, j = size - 1; i < size/2; i++ , j--){
        if(arr[i] == arr[j]){
            flag = 0;
        }
        else{
            flag = 1;
            break;
        }
    }

    if(flag == 0){
        printf("Palindrome");
    }
}

```

```

    }
    else{
        printf("Not Palindrome");
    }
}

```

// Q11

```

void ReverseArray(int arr[] , int size){
    int i,j;
    for(i = 0 , j = size - 1 ; i < j ; i++ , j--){
        int temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;
    }
}

```

// Q12

```

int NearestLesser(int arr[] , int size , int target){
    int lesser = INT_MIN;
    for(int i = 0 ; i < size ; i++){
        if(arr[i] < target && arr[i] > lesser){
            lesser = arr[i];
        }
    }
    return lesser;
}

```

```

int NearestGreater(int arr[] , int size , int target){
    int greater = target;
    for(int i = 0 ; i < size ; i++){
        if(arr[i] > target){

```

```

        greater = arr[i];
    }
}
return greater;
}

```

// Q13

```

void SortMixArray(int A[] , int B[] , int C[] ){
    int i,j = 0 ,k = 0;
    for(i = 0 ; i < 10 ; i++){
        if(j < 5 && (k >= 5 || A[j] <= B[k])){
            C[i] = A[j];
            j++;
        }
        else{
            C[i] = B[k];
            k++;
        }
    }
}

```

// Q14

```

void GenerateArray(int arr[]){
    int i, n;
    for (i = 1; i < 1000; i++) {
        n = rand() % (1000 - 1 + 1) + 1;
        arr[i] = n;
    }
}

int NewArray(int arr[] , int newarr[]){
    int size = 0;

```



```
for(int i = 0 ; i < 1000 ; i++){  
    if((arr[i] % 8 == 0) || (arr[i] % 15 == 0)){  
        newarr[size] = arr[i];  
        size++;  
    }  
}  
return size;  
}
```

// Q15

```
int SecondLargest(int arr[] , int size){  
    int max = INT_MIN;  
    int smax = INT_MIN;  
    for(int i = 0 ; i < size ; i++){  
        if(arr[i] > max){  
            smax = max;  
            max = arr[i];  
        }  
        else if(smax < arr[i] && max != arr[i]){  
            smax = arr[i];  
        }  
    }  
    return smax;  
}
```